TrES-3b

R-0380

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_250622` · created 2026-07-18T19:56:07+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460848.7654 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.03
Transit depth (Rp/Rs)^2	3.81 +/- 1.19 [%]
Orbital Inclination (inc)	81.05 +/- 0.73
Airmass coefficient 1 (a1)	0.889 +/- 0.012
Airmass coefficient 2 (a2)	0.11 +/- 0.035
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#1 - [460, 391]
Optimal Method	PSF photometry
Transit Duration (day)	0.05 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.03 vs 0.16309 (deviation 0.93σ)
Duration fit vs published	0.05 d vs 0.0567 d (deviation 0.58σ)
Mid-time O–C	-3.7 min
Depth SNR	5.7
Residual scatter	1.35 %

Light curve

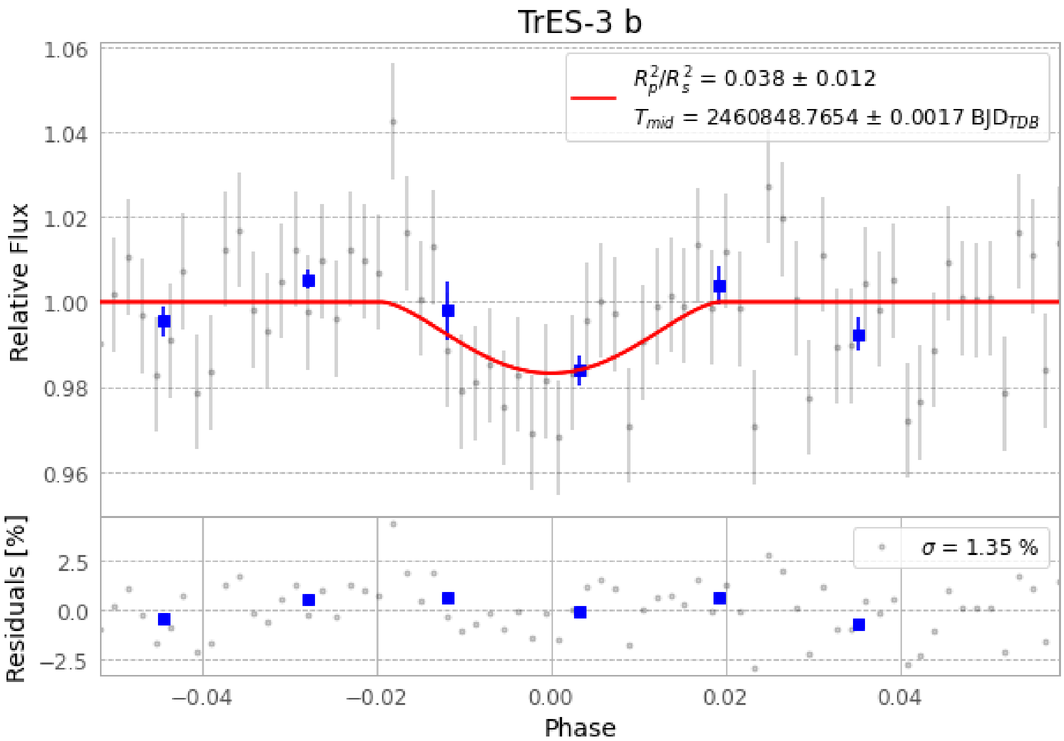

Figure 1 — FinalLightCurve_TrES-3 b_2025-06-21.png (SHA-256 14fe39f1ac7fea14…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 233], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 eb156ac1fc139d30…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b`

Data provenance

70 FITS frames (46 MB), `TRES-3250622043915.FITS` → `TRES-3250622080616.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250622.log (ed7eacc51c61…), FinalLightCurve_TrES-3 b_2025-06-21.png (14fe39f1ac7f…), FinalLightCurve_TrES-3 b_2025-06-21.csv (05ab9ace703c…)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)